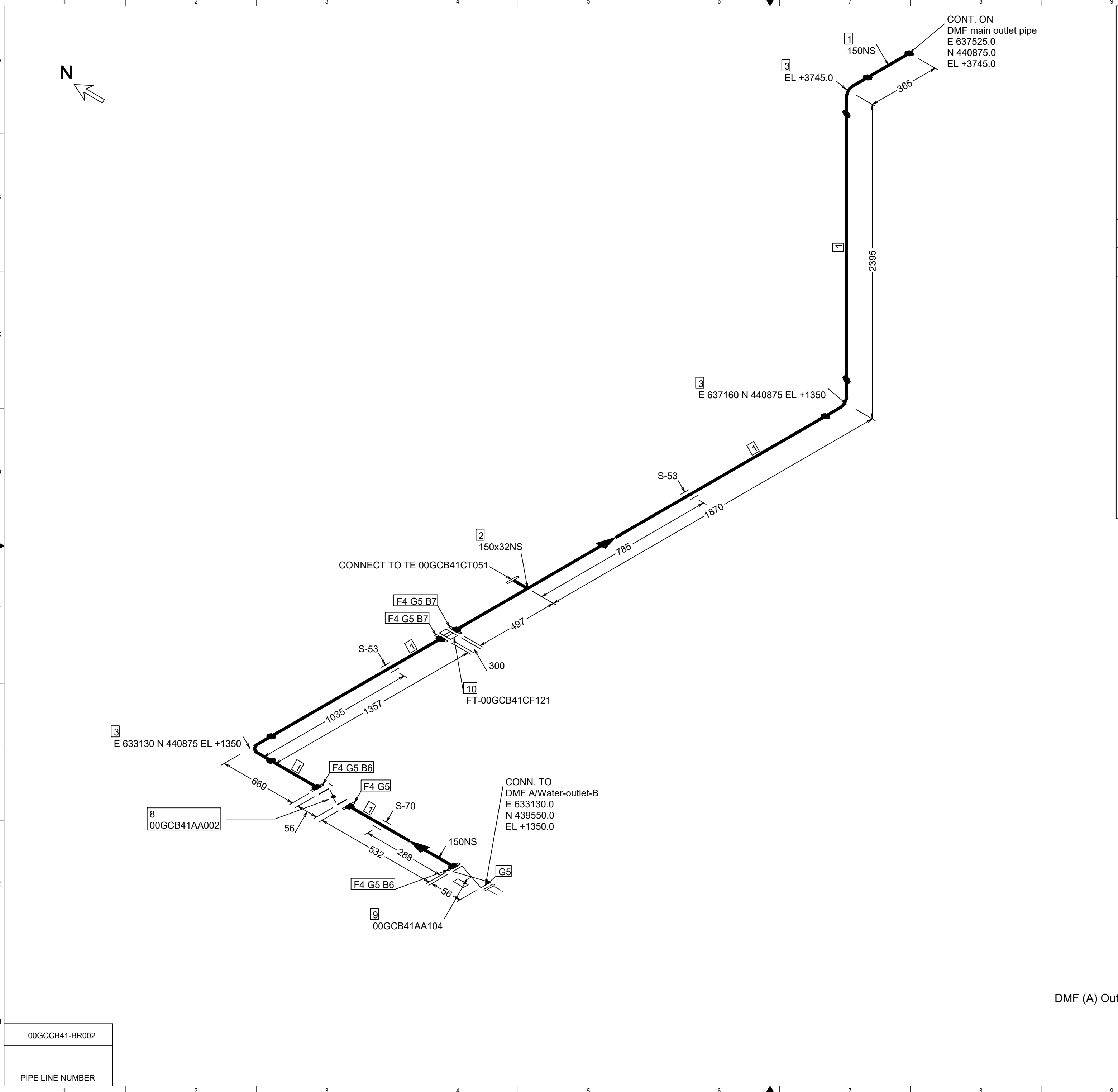


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FABRICATION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
1	PIPE	150	GRP	5934(MM)
	DN150 PIPE, SPECIAL, Class150			
2	PIPING ELEMENT	150	GRP	1
	DN32 FLANGE SHORT TUBE FOR INSTRUMENT, Class150			
	DN150 90° ELBOW, SPECIAL, Class150			
4	FLANGES	150	GRP	5
	DN150 FLANGE, Class150			

ERECTION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
5	GASKETS	150	EPDM	6
	DN150 GASKET, Class150			
6	BOLTS	20	DACROMET	16
	145 mm STUD/NUT/WASHER			
7	85 mm STUD/NUT/WASHER	20	DACROMET	16
8	VALVES / IN-LINE ITEMS	150	CS/EPDM	1
	DN150 MANUAL CLAMP BUTTERFLY VALVE, Class150			
	DN150 PNEUMATIC CLAMP BUTTERFLY VALVE, Class150 00GCB41AA104			
10	INSTRUMENTS	150	GRP	1
	DN150 FLOW TRANSMITTER, Class150 FT-00GCB41CF121			

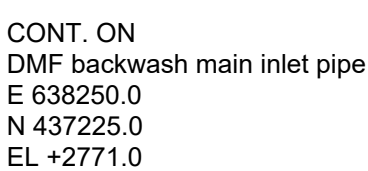
DMF (A) Outlet Pipe

0C	12-05-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
0B	12-03-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
0A	14-02-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
REV.	DATE	DESCRIPTION		BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCPP EXPANSION PROJECT							
PRINCIPAL BUYER:		 SAUDI POWER PROCUREMENT COMPANY							
PROJECT COMPANY:		 SAUDI ELECTRICITY COMPANY							
PROJECT COMPANY'S CONSULTANT:		FICHTNER FICHTNER GMBH & Co. KG							
EPC CONTRACTOR CONSORTIUM:		  SEPCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY							
SEPCO III CONSULTANT:		 SARGENT & LUNDY							
SEPCO IIICONSULTANT:		 JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD							
	DATE	NAME	TITLE:						
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP						
DESIGNED	12-05-26	ZYH							
CHECKED	12-05-26	LB							
APPROVED	12-05-26	DHX	INTERNAL DOCUMENT NO.:			DOCUMENT OWNER: SARGENT & LUNDY			
DEPARTMENT- MECHANICAL									
DRG. NO.: PP12EXP-P109-G_-MDA-07895				SCALE: NA		SHT: 1 OF 453		REV. 0C	



0C	12-05-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
0B	12-03-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
0A	14-02-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
REV.	DATE	DESCRIPTION			BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCPP EXPANSION PROJECT								
PRINCIPAL BUYER:				SAUDI POWER PROCUREMENT COMPANY						
PROJECT COMPANY:				SAUDI ELECTRICITY COMPANY						
PROJECT COMPANY'S CONSULTANT:		FICHTNER		FICHTNER GMBH & Co. KG						
EPC CONTRACTOR CONSORTIUM:				SEPCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY						
SEPCO III CONSULTANT:				SARGENT & LUNDY						
SEPCO IIICONSULTANT:				JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD						
	DATE	NAME	TITLE:							
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP							
DESIGNED	12-05-26	ZYH								
CHECKED	12-05-26	LB								
APPROVED	12-05-26	DHX								
DEPARTMENT:- MECHANICAL			INTERNAL DOCUMENT NO.:				DOCUMENT OWNER: SARGENT & LUNDY			
DRG. NO.: PP12EXP-P109-G _MDA-07895					SCALE : NA		SHT: 2 OF 453		REV. OC	
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FILE NAME: Piping Isometric drawings for WTP&ETP										

FILE NAME: Piping Isometric drawings for WTP&ETP

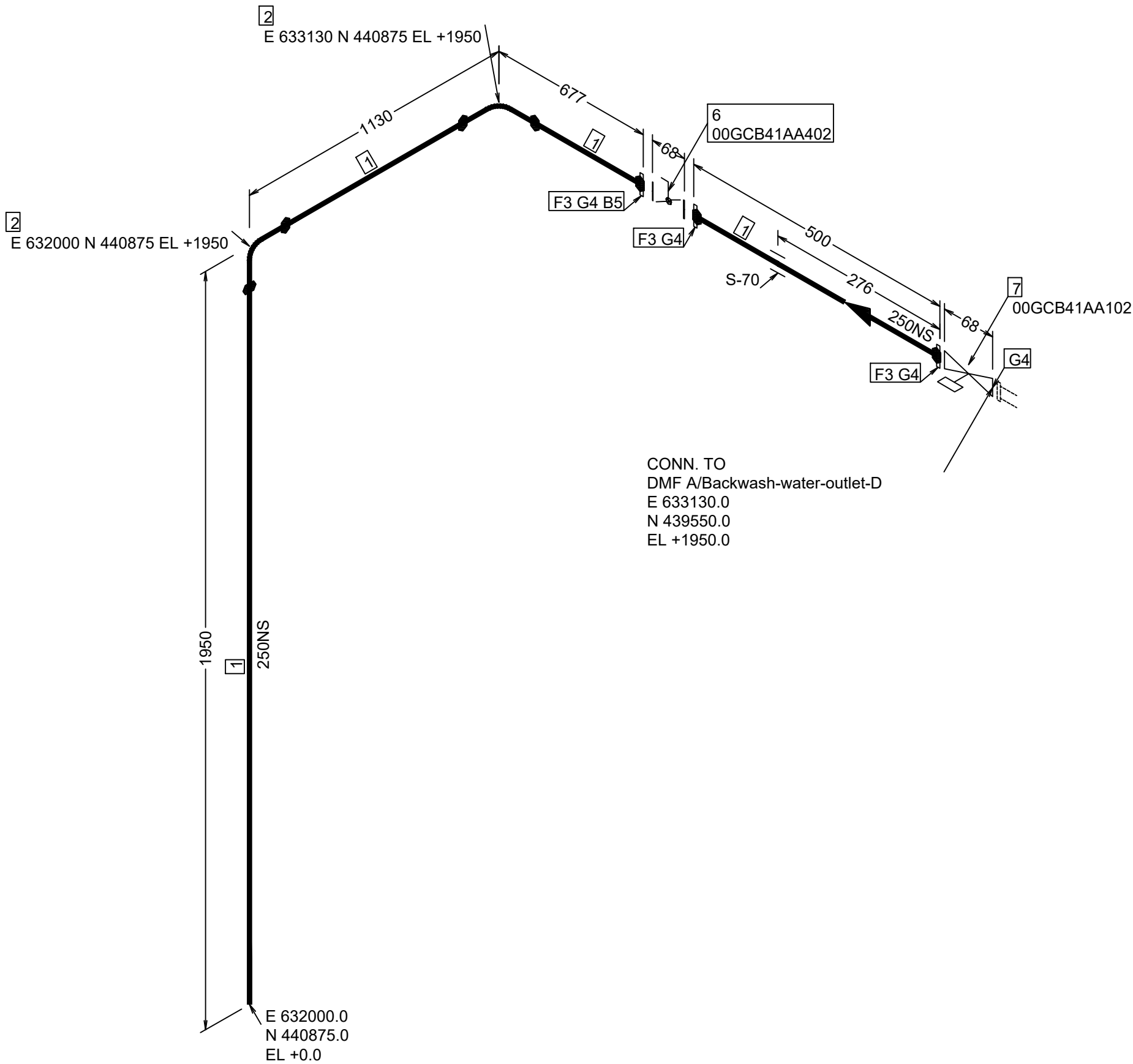


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0B	12-03-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
0A	14-02-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
REV.	DATE	DESCRIPTION			BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCPP EXPANSION PROJECT								
PRINCIPAL BUYER:		SAUDI POWER PROCUREMENT COMPANY 								
PROJECT COMPANY:		SAUDI ELECTRICITY COMPANY 								
PROJECT COMPANY'S CONSULTANT:		FICHTNER FICHTNER GMBH & Co. KG FICHTNER								
EPC CONTRACTOR CONSORTIUM:		SEPCHO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY  								
SEPCHO III CONSULTANT:		SARGENT & LUNDY 								
SEPCHO IICONCONSULTANT:		JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD 								
	DATE	NAME	TITLE:							
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP							
DESIGNED	12-05-26	ZYH								
CHECKED	12-05-26	LB								
APPROVED	12-05-26	DHX	INTERNAL DOCUMENT NO.:				DOCUMENT OWNER: SARGENT & LUNDY			
DEPARTMENT:- MECHANICAL										
DRG. NO.: PP12EXP-P109-G _MDA-07895					SCALE : NA		SHT: 3 OF 453		REV. 0C	

DMF (A) Backwash Inlet Pipe

FILE NAME: Piping Isometric drawings for WTP&ETP

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CONN. TO
DMF A/Backwash-water-outlet-D
E 633130.0
N 439550.0
EL +1950.0

FABRICATION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
1	<u>PIPE</u> DN250 PIPE, SPECIAL, Class150	250	GRP	2517(MM)
	<u>PIPING ELEMENT</u> DN250 90° ELBOW, SPECIAL, Class150			
2	<u>FLANGES</u> DN250 FLANGE, Class150	250	GRP	2
3		250	GRP	3
ERECTION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
4	<u>GASKETS</u> DN250 GASKET, Class150	250	EPDM	4
	<u>BOLTS</u> 175 mm STUD/NUT/WASHER			
5	<u>VALVES / IN-LINE ITEMS</u> DN250 MANUAL CLAMP BUTTERFLY VALVE, Class150	20	DACROMET	12
6		250	CS/EPDM	1
7		250	CS/EPDM	1

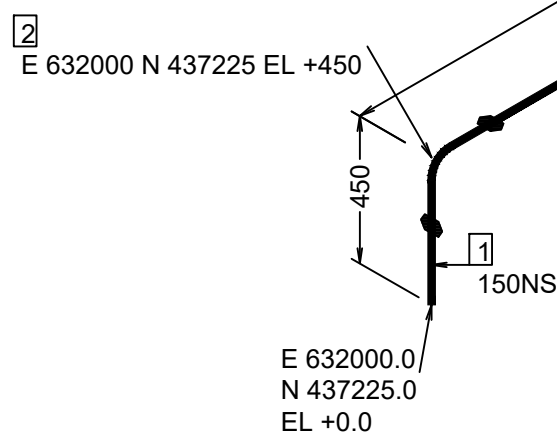
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0B	12-03-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX	
0A	14-02-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX	
REV.	DATE	DESCRIPTION			BY	MECH.	ELEC.	CIVIL	C&I	APPD.	
PROJECT :		PP12 CCPP EXPANSION PROJECT									
PRINCIPAL BUYER:		 SAUDI POWER PROCUREMENT COMPANY									
PROJECT COMPANY:		 SAUDI ELECTRICITY COMPANY									
PROJECT COMPANY'S CONSULTANT:		 FICHTNER GMBH & Co. KG									
EPC CONTRACTOR CONSORTIUM:		  SEPCO III ELECTRIC POWER CONSTRUCTION CO. LTD.  DOOSAN ENERBILITY									
SEPCO III CONSULTANT:		 SARGENT & LUNDY									
SEPCO III CONSULTANT:		 JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD									
		DATE	NAME	TITLE:							
PREPARED		12-05-26	CJC	Piping Isometric drawings for WTP&ETP							
DESIGNED		12-05-26	ZYH								
CHECKED		12-05-26	LB								
APPROVED		12-05-26	DHX								
DEPARTMENT:- MECHANICAL				INTERNAL DOCUMENT NO.:			DOCUMENT OWNER: SARGENT & LUNDY				
DRG. NO.: PP12EXP-P109-G__MDA-07895					SCALE : NA		SHT: 4 OF 453		REV. 0C		
10					11					12	
FILE NAME: Piping Isometric drawings for WTP&ETP											

FILE NAME: Piping Isometric drawings for WTP&ETP

DMF (A) Backwash Drainage Pipe

00GCB41-BR402

PIPE LINE NUMBER

[illegible]

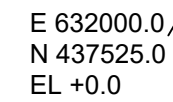
DMF (A) Drainage Pipe



0C	12-05-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX		
0B	12-03-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX		
0A	14-02-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX		
REV.	DATE	DESCRIPTION	BY	MECH.	ELEC.	CIVIL	C&I	APPD.		
PP12 CCPP EXPANSION PROJECT										
PROJECT :										
PRINCIPAL BUYER:		SAUDI POWER PROCUREMENT COMPANY 								
PROJECT COMPANY:		SAUDI ELECTRICITY COMPANY 								
PROJECT COMPANY'S CONSULTANT:		FICHTNER FICHTNER GMBH & Co. KG FICHTNER								
EPC CONTRACTOR CONSORTIUM:		SEPSCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENENERBIT 								
SEPSCO III CONSULTANT:		SARGENT & LUNDY 								
SEPSCO IIconSULTANT:		JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD 								
	DATE	NAME	TITLE:							
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP							
DESIGNED	12-05-26	ZYH								
CHECKED	12-05-26	LB								
APPROVED	12-05-26	DHX								
DEPARTMENT - MECHANICAL			INTERNAL DOCUMENT NO.:			DOCUMENT OWNER: SARGENT & LUNDY				
DRG. NO.: PP12EXP-P109-G _MDA-07895				SCALE : NA		SHT: 6 OF 453		REV. 0C		

DMF (A) Inlet pipe

FILE NAME: Piping Isometric drawings for WTP&ETP

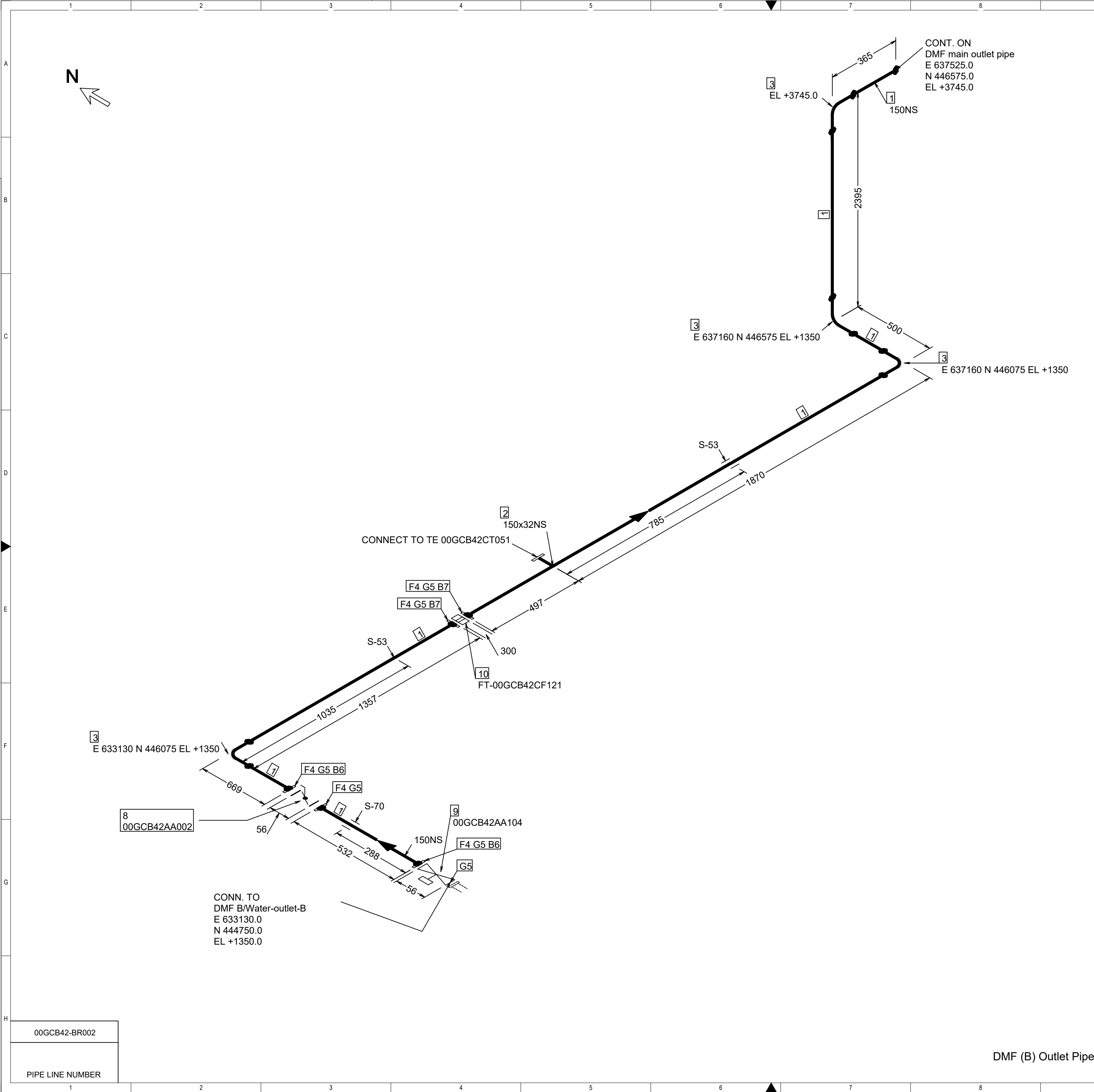


00GCB41-BR501

PIPE LINE NUMBER

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0B	12-03-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
0A	14-02-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
REV.	DATE	DESCRIPTION			BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCPP EXPANSION PROJECT								
PRINCIPAL BUYER:				SAUDI POWER PROCUREMENT COMPANY						
PROJECT COMPANY:				SAUDI ELECTRICITY COMPANY						
PROJECT COMPANY'S CONSULTANT:		FICHTNER		FICHTNER GMBH & Co. KG						
EPC CONTRACTOR CONSORTIUM:				SEPCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY						
SEPCO III CONSULTANT:				SARGENT & LUNDY						
SEPCO IIICONSULTANT:				JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD						
	DATE	NOME	TITLE:							
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP							
DESIGNED	12-05-26	ZYH								
CHECKED	12-05-26	LB								
APPROVED	12-05-26	DHX								
DEPARTMENT:- MECHANICAL			INTERNAL DOCUMENT NO.:				DOCUMENT OWNER: SARGENT & LUNDY			
DRG. NO.: PP12EXP-P109-G _MDA-07895					SCALE : NA		SHT: 7 OF 453		REV. 0C	

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FABRICATION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
1	PIPE	150	GRP	5984(MM)
	DN150 PIPE, SPECIAL, Class150			
2	PIPING ELEMENT	150	GRP	1
	DN32 FLANGE SHORT TUBE FOR INSTRUMENT, Class150			
	DN150 90° ELBOW, SPECIAL, Class150			
4	FLANGES	150	GRP	5
	DN150 FLANGE, Class150			

ERECTION MATERIALS				
PT NO.	DESCRIPTION	DN (MM)	MATERIAL	QTY.
5	GASKETS	150	EPDM	6
	DN150 GASKET, Class150			
6	BOLTS	20	DACROMET	16
	145 mm STUD/NUT/WASHER			
7		20	DACROMET	16
	85 mm STUD/NUT/WASHER			
8	VALVES / IN-LINE ITEMS	150	CS/EPDM	1
	DN150 MANUAL CLAMP BUTTERFLY VALVE, Class150			
	DN150 PNEUMATIC CLAMP BUTTERFLY VALVE, Class150 00GCB42AA104			
10	INSTRUMENTS	150	GRP	1
	DN150 FLOW TRANSMITTER, Class150 FT-00GCB42CF121			

0C	12-05-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
0B	12-03-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
0A	14-02-26	ISSUED FOR APPROVAL		ZYH	WJY	ZXJ	DXR	ZYH	DHX
REV.	DATE	DESCRIPTION		BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCBP EXPANSION PROJECT							
PRINCIPAL BUYER:		SAUDI POWER PROCUREMENT COMPANY							
PROJECT COMPANY:		SAUDI ELECTRICITY COMPANY							
PROJECT COMPANY'S CONSULTANT:		FICHTNER FICHTNER GMBH & Co. KG							
EPC CONTRACTOR CONSORTIUM:		SEPCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY							
SEPCO III CONSULTANT:		SARGENT & LUNDY							
SEPCO III CONSULTANT:		JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD							
	DATE	NAME	TITLE:						
PREPARED	12-05-26	CJC	Piping Isometric drawings for WTP&ETP						
DESIGNED	12-05-26	ZYH							
CHECKED	12-05-26	LB							
APPROVED	12-05-26	DHX	INTERNAL DOCUMENT NO.:		DOCUMENT OWNER: SARGENT & LUNDY				
DEPARTMENT:- MECHANICAL									
DRG. NO.: PP12EXP-P109-G_-MDA-07895				SCALE: NA	SHT: 8 OF 453		REV. 0C		
10		11		12					

FILE NAME: Piping Isometric drawings for WTP&ETP



0C	12-05-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX	
0B	12-03-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX	
0A	14-02-26	ISSUED FOR APPROVAL	ZYH	WJY	ZXJ	DXR	ZYH	DHX	
REV.	DATE	DESCRIPTION	BY	MECH.	ELEC.	CIVIL	C&I	APPD.	
PROJECT :		PP12 CCPP EXPANSION PROJECT							
PRINCIPAL BUYER:		 SAUDI POWER PROCUREMENT COMPANY							
PROJECT COMPANY:		 SAUDI ELECTRICITY COMPANY							
PROJECT COMPANY'S CONSULTANT:		FICHTNER FICHTNER GMBH & Co. KG							
EPC CONTRACTOR CONSORTIUM:		  DOOSAN DOOSAN ENERBILITY							
SEPCO III CONSULTANT:		 SARGENT & LUNDY							
SEPCO IIconSULTANT:		 JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD							
		DATE	NAME	TITLE:					
PREPARED		12-05-26	CJC	Piping Isometric drawings for WTP&ETP INTERNAL DOCUMENT NO.: DEPARTMENT:- MECHANICAL	DOCUMENT OWNER: SARGENT & LUNDY				
DESIGNED		12-05-26	ZYH						
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DRG. NO.: PP12EXP-P109-G_-MDA-07895				SCALE : NA	SHT: 9 OF 453		REV. 0C		
FILE NAME: Piping Isometric drawings for WTP&ETP			11			12			

FILE NAME: Piping Isometric drawings for WTP&ETP



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0B	12-03-26	ISSUED FOR APPROVAL			ZYH	WJY	ZXJ	DXR	ZYH	DHX
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REV.	DATE	DESCRIPTION			BY	MECH.	ELEC.	CIVIL	C&I	APPD.
PROJECT :		PP12 CCPP EXPANSION PROJECT								
PRINCIPAL BUYER:				SAUDI POWER PROCUREMENT COMPANY						
PROJECT COMPANY:				SAUDI ELECTRICITY COMPANY						
PROJECT COMPANY'S CONSULTANT:		FICHTNER		FICHTNER GMBH & Co. KG						
EPC CONTRACTOR CONSORTIUM:				SEPSCO III ELECTRIC POWER CONSTRUCTION CO. LTD. DOOSAN ENERBILITY						
SEPCO III CONSULTANT:				SARGENT & LUNDY						
SEPCO IIICONSULTANT:				JIANGSU TIANMA E. P. TECHNOLOGY GROUP CO., LTD						
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FILE NAME: Piping Isometric drawings for WTP&ETP